

Allocation:

Our group consists of two members majoring in computer science-related disciplines and another two members who have completed coursework in Bitcoin and blockchain-related subjects. Based on this complementary background, we selected the decentralized cryptocurrency market as our primary asset class for analysis. Considering our limited capital constraints as students, we adopt an hourly trading window to reduce transaction frequency and associated trading costs, while maintaining sufficient data granularity for arbitrage opportunity detection.

Abstract:

This project investigates cross-pool MEV arbitrage in decentralised exchange (DEX) liquidity pools. The strategy tests whether temporary price differences across pools quoting the same asset pair can be converted into executable profit after accounting for slippage, DEX fees, and gas costs.

The topic was selected because it is more specialised than a standard directional trading strategy and is closely related to market microstructure. In automated market makers (AMMs), prices are determined by pool reserves rather than an order book, which can create temporary cross-pool price discrepancies. However, these discrepancies are only meaningful if they remain profitable after realistic execution frictions are included. Therefore, the objective of this project is not merely to identify quoted price gaps, but to test whether they can support a robust and executable arbitrage strategy. After Validation and backtesting, the strategy did not work in various assets after considering the cost. Thus, we change the strategy from arbitrage to price prediction.

Data Processing:

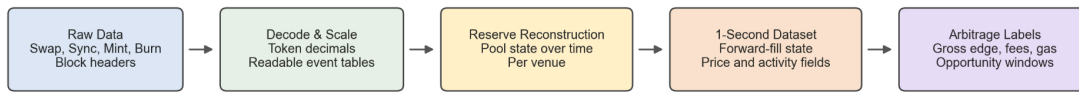
The project reconstructs historical pool states from on-chain event data. For each selected pool, we track reserve changes through time, align the reserves to a regular frequency, derive the implied AMM price, and then simulate a hypothetical arbitrage trade between venues.

This methodology matters because simply comparing displayed prices is not enough. In an AMM, the execution price depends on trade size. A quoted gap may disappear once the trade actually moves the pool. For that reason, the project required a full preprocessing and data-engineering pipeline rather than a simple price-comparison script.

Preprocessing And Data Processing Pipeline:

After raw collection, the data passes through several preprocessing steps before it can be used for arbitrage analysis.

Preprocessing and Data-Processing Pipeline



The project transforms raw on-chain pool events into second-level executable arbitrage labels, then summarizes the results with size-sensitivity tables and report figures.

The main processing steps are:

1. Decode event payloads: convert raw log data into readable event tables.
2. Scale token amounts: convert integer blockchain units into real WBTC, WETH, USDC, and other token quantities using token decimals.
3. Reconstruct reserves: rebuild pool reserves over time from the event sequence.
4. Resample to regular frequency: align the data to a fixed time grid, especially the 1-second grid used in the second-level backtests.
5. Forward-fill state between updates: carry forward the most recent reserve state until the next on-chain reserve update.
6. Create derived fields: compute mid-prices, swap counts, trading volume, liquidity snapshots, and gas conversion fields.
7. Construct arbitrage labels: simulate hypothetical trades across venues and calculate gross edge, fees, gas, and final net profitability.
8. Group positive runs into windows: convert consecutive positive seconds into opportunity windows so that one persistent state is not misread as many independent trades.

This preprocessing pipeline is central to the project. Without reserve reconstruction and time alignment, it would not be possible to simulate an executable trade correctly.

Strategy Selection:

Initial Arbitrage Test: 1-Minute Screening:

The first stage of the project used a 30-day, 1-minute screening design for WBTC/WETH. This stage was intended to determine whether the pair displayed any clear arbitrage signal before a more detailed execution dataset was built. The results did not show a convincing actionable net-profitable strategy.

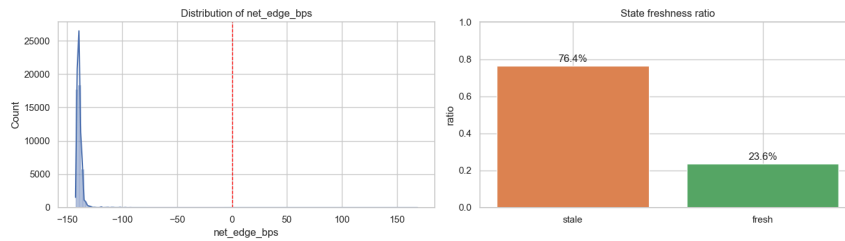


Figure 1: Net edges are almost always negative

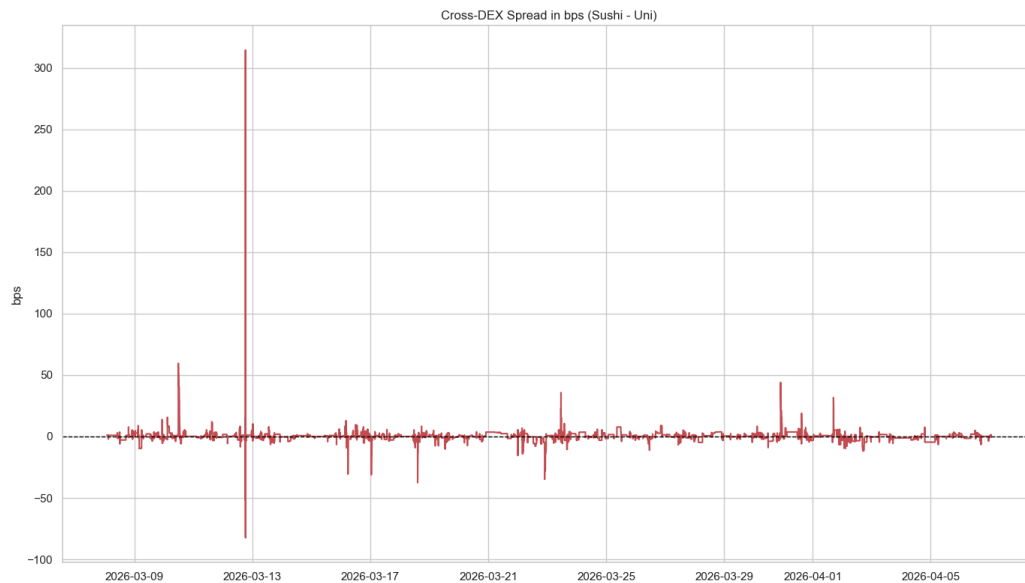


Figure 2: Cross DEX Spread are generally too small to arbitrage

One likely explanation is that the 1-minute interval was too coarse for arbitrage analysis. In DEX markets, temporary price discrepancies may be corrected within seconds. A 1-minute frequency may therefore fail to capture the microstructure dynamics needed to assess whether an opportunity is genuinely executable. This limitation motivated the move to a higher-frequency framework.

Refined Arbitrage Test: 1-Second Execution Analysis:

To address the limitation of the first-stage screen, the analysis was refined to a 7-day, 1-second reserve-based execution test for WBTC/WETH on Optimism. This stage modelled arbitrage more realistically by reconstructing pool states from on-chain reserve events and simulating hypothetical trades after accounting for slippage, pool fees, and gas assumptions.

Although the higher-frequency dataset revealed many more short-lived price differences, the overall conclusion did not change. Gross-positive opportunities declined rapidly as trade size increased, while net-positive seconds remained zero at all tested sizes. In the saved Optimism WBTC/WETH execution run, the project recorded zero positive seconds and zero positive opportunity windows after costs were included. This means that no executable net-profitable arbitrage trade was identified in the tested sample.

Broader Market Evidence:

After the WBTC/WETH tests failed to produce a convincing strategy, the project expanded to multiple chains, exchanges, and pairs. The broader comparison showed a consistent pattern: raw spreads often appeared at the quote level, and some opportunities survived explicit fee adjustment, but almost none remained after stricter fee-and-gas assumptions were imposed.

The exploratory Solana extension produced more candidate windows under lower-bound assumptions, but those opportunities were economically weak because the pools were very small and the estimated profits were minimal. As a result, these findings could not support a robust scalable trading claim.

Why the Arbitrage Strategy Failed:

The evidence suggests that the arbitrage strategy failed for two main reasons. First, the initial 1-minute screen was too coarse to capture very short-lived market discrepancies reliably. Second, even after moving to a 1-second reserve-based execution framework, the apparent opportunities did not survive realistic trading frictions. Smaller trades reduced slippage, but they also reduced gross profit, while DEX fees and gas costs remained material. As a result, the underlying economic edge was too weak to produce robust net profitability.

The central conclusion is therefore that quoted inefficiency does not imply executable profit. In the tested DEX environments, many attractive-looking arbitrage opportunities disappeared once slippage, swap fees, and gas costs were incorporated properly.

Transition to Price Prediction:

These findings motivate a shift from reactive arbitrage to a predictive framework. The earlier analysis suggests that waiting for an obvious observed spread is often too late, because the remaining edge is insufficient after costs. The revised research direction is therefore to examine whether short-term price movements can be predicted before the apparent opportunity is fully arbitrated away. In this sense, price prediction is not a separate topic, but a refinement of the original trading objective.

Interim Conclusion:

The project has not produced a convincing live-tradable arbitrage strategy. The 1-minute screening stage was too coarse, and the refined 1-second execution test still showed no net-profitable arbitrage after realistic costs were included. The broader market comparison reinforced the same conclusion. However, this is still a valuable research result, because it shows that execution realism matters more than raw opportunity counts in DEX arbitrage. This finding provides a clear basis for the next stage of the project, namely the move toward price prediction.

Price Prediction as the Refined Strategy:

Because the reactive arbitrage framework did not produce robust net-profitable results, the project shifted toward a predictive framework. The earlier evidence suggested that waiting for an already visible cross-pool spread was often too late, since the remaining edge was usually too small to survive slippage, DEX fees, and gas costs. This refinement is consistent with the project requirements, which allow model improvement through price prediction and other advanced methods.

Instead of reacting to an observed arbitrage gap, the revised objective is to predict short-term price movement and use that forecast to generate a directional trading signal. In this sense, price prediction is not a separate topic, but a refinement of the original research aim: improving trade timing and signal quality in an environment where simple spread-based arbitrage proved unreliable. This direction also fits the course guidance on market microstructure topics and intraday price prediction models.

Data and Feature Design for Price Prediction:

The price prediction model uses hourly market data from the WETH/USDC setting and combines price-based and microstructure-based variables. The feature set includes log returns, return acceleration, volume-weighted spread, RSI, and observed spread in basis points. These variables are designed to capture both short-term momentum and trading-condition information, rather than relying only on past returns. This is consistent with the assignment expectation that background analysis should inform the design of the strategy and model.

The target variable is the next-period return. The modelling choice therefore reframes the project from identifying realised arbitrage windows to forecasting whether the market is likely to move in a favourable direction in the following period. Compared with the failed arbitrage framework, this approach is more forward-looking and is better aligned with the idea that market inefficiencies may disappear too quickly once they become visible.

Model Development:

The refined model is a multi-factor LSTM designed to capture temporal dependence in sequential market data. The model uses a 24-period rolling input window and applies an 80/20 split between training and testing samples. Before model construction, both inputs and

targets are standardised using only the training portion of the sample. This follows the assignment requirement to separate model development from out-of-sample validation.

The use of an LSTM is appropriate because the project no longer focuses on a static spread condition at a single timestamp. Instead, the goal is to learn whether recent sequences of returns, acceleration, spreads, and momentum indicators contain information about the next-period move. In that sense, the refined model extends the earlier analysis rather than replacing it: the original arbitrage tests showed that visible opportunities were not sufficient, while the prediction model attempts to extract information before the opportunity is fully priced away.

Testing and Preliminary Backtest:

After generating out-of-sample predictions, we applied a directional trading rule. A long signal is taken only when the predicted return exceeds a confidence threshold linked to transaction costs. Net returns are then calculated after a fee assumption, and the strategy is compared with a buy-and-hold benchmark.

The preliminary backtest suggests that the refined framework is more promising than the original arbitrage strategy. In the uploaded notebook, the final strategy return is reported as 4.18%, while the strategy is in the market only 18.71% of the time. Although this result should still be treated as preliminary, it indicates that the predictive framework may generate a more selective and economically meaningful trading rule than the earlier reactive arbitrage design.

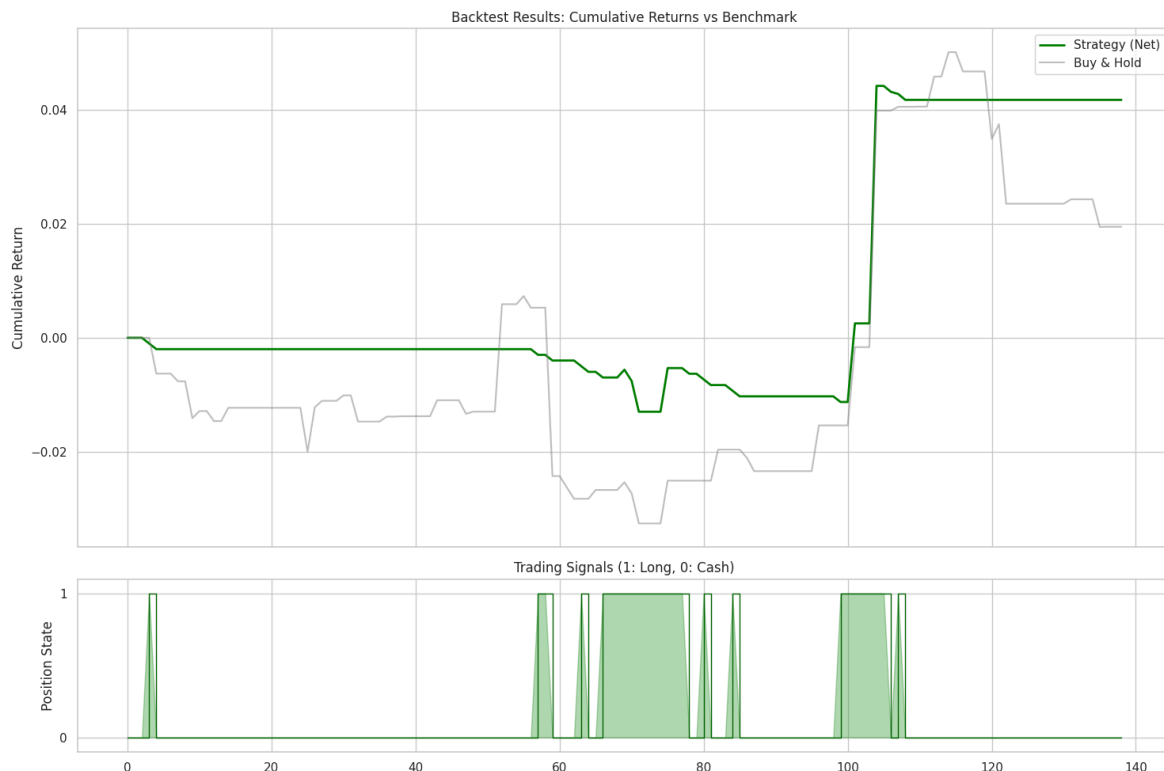


Figure 3: Backtesting Trading Strategy

Interpretation

The refined price prediction framework changes the role of market microstructure information in the project. In the original arbitrage stage, spreads were treated as immediate trading opportunities. In the revised framework, similar market information is treated as an input for forecasting. This is an important conceptual improvement. The earlier findings showed that observed spreads often disappeared after execution costs were included, whereas the prediction model attempts to anticipate price adjustment before the remaining edge is fully eroded.

However, this section should still be presented carefully. The current result is a model-based backtest rather than proof of a live-tradable production strategy. The final report should therefore emphasise that the LSTM results are a refinement stage following the rejection of the initial arbitrage hypothesis, rather than a definitive claim of strong market inefficiency.

Conclusion:

The overall project now supports a two-stage conclusion. First, the cross-pool arbitrage hypothesis was tested and rejected under realistic execution assumptions. The 1-minute screening stage was too coarse, and the later 1-second execution analysis still found no net-profitable arbitrage once slippage, fees, and gas costs were included.

Second, this failure motivated a refinement toward short-term price prediction. The uploaded LSTM model uses sequential return and microstructure features to predict next-period movement and produces encouraging preliminary backtest results. Therefore, the project evolves from testing whether visible arbitrage gaps can be captured to testing whether predictive signals can improve trading timing in markets where simple arbitrage is not robust. This progression is more defensible than presenting the price prediction model as an unrelated new topic, because it is directly motivated by the empirical failure of the first strategy.

Critical Reflection and Real-World Implications:

A major strength of this project is that it evaluates trading opportunities under realistic execution assumptions rather than relying only on quoted price differences. This is important in real-world trading because an apparent arbitrage spread is not economically meaningful unless it remains profitable after slippage, DEX fees, gas costs, and liquidity constraints are taken into account. The project therefore produces a practical result: visible cross-pool mispricing does not automatically imply executable profit.

The project also has several limitations. The initial 1-minute screening framework was too coarse to capture very short-lived market discrepancies, while later extensions across markets were not always perfectly comparable because liquidity, fee structure, and execution assumptions differed across chains and pools. In addition, even the refined price prediction stage should still be treated cautiously, since a backtest result does not guarantee that the strategy would remain profitable under live execution conditions. These limitations matter in practice because real-world trading performance depends not only on model quality, but also on latency, transaction-cost stability, and market regime changes.

From a real-world perspective, the most important lesson from the project is that execution realism matters more than raw opportunity counts. For arbitrage strategies, this means that traders should prioritise executable net profit at realistic size rather than the number of observed spreads. For predictive strategies, this means that model performance should be evaluated together with trading rules, costs, and implementation feasibility. The transition from reactive arbitrage to price prediction is therefore practically meaningful, because it reflects a move from identifying visible inefficiencies to anticipating short-term market movement before the opportunity is fully priced away.

Recommendations for Real-World Implementation:

First, future work should test deeper and more comparable markets. In practice, a strategy is only useful if it can be executed at meaningful scale, so shallow pools and highly fragmented venues should not be treated as sufficient evidence of deployable profitability.

Second, execution-cost modelling should be improved further. A real trading strategy must account for changing gas fees, fee tiers, slippage, and liquidity conditions in a more precise way, since these factors determine whether a theoretical edge can be converted into realised profit.

Third, the final model evaluation should focus on implementation outcomes rather than model outputs alone. In a real-world setting, the relevant question is not only whether the model predicts correctly, but whether it improves trade timing and remains profitable after costs, risk controls, and operational constraints are applied.